Jayakumar M

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PROFESSIONAL SUMMARY

Junior Software Engineer developing C#/.NET and WPF/XAML software for medical robotics, with AI/ML healthcare project experience and knowledge of ISO 13485:2016, internal auditing, CAPA, and risk assessment.

TECHNICAL SKILLS

Programming Languages: Python, C#, Java, SQL, C, C++
Software Development: .NET, WPF, XAML, MVVM, Object-Oriented Programming, Debugging, Exception Handling, Data Structures and Algorithms
AI and Machine Learning: Supervised Learning, Classification, Regression, CNNs, Computer Vision, Image Processing, Image Segmentation, Data Preprocessing, Feature Engineering, Model Evaluation, Confusion Matrix, Classification Report
Libraries and Tools: NumPy, Pandas, scikit-learn, TensorFlow, PyTorch, OpenCV, Matplotlib, Git, GitHub, Azure, Bugzilla, SVN
Databases and Backend: SQL Server, SSMS, MySQL, REST API Concepts, Data Processing, CRUD Operations
Quality and Compliance: ISO 13485:2016 QMS, ISO 19011:2026 Internal Audit, Corrective and Preventive Action (CAPA), Risk Assessment and Control, Audit Documentation

WORK EXPERIENCE

Junior Software Engineer
June 2025 – Present
Perfint Healthcare Pvt Ltd
Chennai, Tamil Nadu, India

- Develop C#/.NET backend functionality and WPF/XAML interfaces for a medical robotic navigation device used in spine surgery.
- Implement navigation workflows, integrate software components, and use debugging/testing to resolve functional issues; apply AI-assisted coding tools to simplify code and reduce repetitive effort.
- Investigated navigation-accuracy issues during clinical trials and contributed to troubleshooting and workflow improvements with cross-functional teams.
- Review AI-assisted coding suggestions for correctness and maintainability.
- Collaborate with Software, QA, Mechanical, and Manufacturing teams; completed internal audit activities in Production and Sales and apply knowledge of ISO 13485:2016, ISO 19011:2026, CAPA, and risk controls.

PROJECTS

- **Skin Lesion Segmentation and Classification (Python, CNN, OpenCV):** Built a dermoscopic image pipeline using hair/noise removal, inpainting, morphological operations, and LAB-space K-Means segmentation; explored ResNet50 classification and evaluated outputs with confusion matrices and validation curves.
- **Heart Disease Detection using Machine Learning (Python):** Applied preprocessing and feature engineering to a healthcare image-analysis project; used NumPy and Pandas for data handling and evaluated predictive results.
- **Email Client Application (Java, Swing, JavaMail API):** Built a Java Swing desktop email client with JavaMail API for authentication and email server communication.

CERTIFICATIONS AND TRAINING

- Awareness of ISO 13485:2016 Standard and Internal Audit Training based on ISO 19011:2026 Standard – Perfint Healthcare Pvt Ltd.
- Awareness on Implementation of Corrective and Preventive Action – Perfint Healthcare Pvt Ltd.
- Certification in Artificial Intelligence and Machine Learning (Tamil) – Vishlesan i-Hub, IIT Patna.

PUBLICATIONS

- *Heart Disease Detection using Machine Learning Classification Techniques in E-Healthcare Systems* – International Journal of Engineering Research & Technology (IJERT), June 2025.
- *Segmentation and Classification of Hairless Dermoscopic Skin Cancer Images using Deep Learning* – International Journal of Engineering Research & Technology (IJERT), June 2025.

AWARDS AND RECOGNITION

- Best Team of the Year 2025–2026.
- Perfint Night of Town – June 2026.

EDUCATION

Madras Institute of Technology, Anna University
Chennai, Tamil Nadu, India
B.E. Electronics and Communication Engineering — CGPA: 8.44/10
Oct 2021 – Apr 2025